

CS471 – Lesson 18 Solutions Key

Probability: Uncertainty, Conditioning, and Independence

Part 1: Guided In-Class Worksheet (Instructor-Led)

1. Joint Probability Distribution and Marginalization

Scenario: An automated Early Warning Radar monitors an airspace corridor. The system tracks two random variables: Target Presence ($T \in \{\text{Target}, \text{NoTarget}\}$) and Radar Alert status ($A \in \{\text{Alert}, \text{NoAlert}\}$). The full joint probability distribution $P(T, A)$ is given below:

	$A = \text{Alert}$	$A = \text{NoAlert}$
$T = \text{Target}$	0.04	0.01
$T = \text{NoTarget}$	0.05	0.90

- a. **Question:** Verify that this table represents a valid probability distribution.

Solution

All entries are non-negative ($P \geq 0$), and the sum of all joint probabilities across the entire sample space equals exactly 1.0:

$$\sum_{t,a} P(T = t, A = a) = 0.04 + 0.01 + 0.05 + 0.90 = 1.00$$

- b. **Question:** Compute the marginal probability that an actual target is present, $P(T = \text{Target})$, by marginalizing (summing out) over variable A .

Solution

$$P(T = \text{Target}) = \sum_{a \in \{\text{Alert}, \text{NoAlert}\}} P(T = \text{Target}, A = a)$$

$$P(T = \text{Target}) = 0.04 + 0.01 = \mathbf{0.05} \quad (5\%)$$

- c. **Question:** Compute the marginal probability that the radar triggers an alert, $P(A = \text{Alert})$, by marginalizing over variable T .

Solution

$$P(A = \text{Alert}) = \sum_{t \in \{\text{Target}, \text{NoTarget}\}} P(T = t, A = \text{Alert})$$

$$P(A = \text{Alert}) = 0.04 + 0.05 = \mathbf{0.09} \quad (9\%)$$

2. Conditional Probability and Belief Updating

- a. **Question:** Write the mathematical formula for the conditional probability $P(T = \text{Target} \mid A = \text{Alert})$.

Solution

$$P(T = \text{Target} \mid A = \text{Alert}) = \frac{P(T = \text{Target}, A = \text{Alert})}{P(A = \text{Alert})}$$

- b. **Question:** If the radar sounds an alarm ($A = \text{Alert}$), compute the updated probability that an actual enemy target is present. Explain why this value is lower than the radar's raw detection rate.

Solution

$$P(T = \text{Target} \mid A = \text{Alert}) = \frac{0.04}{0.09} = \mathbf{0.444} \quad (\approx 44.4\%)$$

Explanation: Targets are rare in the overall airspace (base rate is only 5%). Even though the radar rarely makes false alarms on empty skies ($P = 0.05$), the vast number of non-target scans means false alarms (0.05) outnumber true detections (0.04), reducing the posterior confidence below 50%.

3. Evaluating Statistical Independence

- a. **Question:** State the mathematical definition of statistical independence between two events A and B .

Solution

Two events A and B are statistically independent ($A \perp B$) if and only if:

$$P(A, B) = P(A)P(B) \quad \text{or equivalently} \quad P(A \mid B) = P(A)$$

- b. **Question:** Using the joint probability table from Section 1, determine whether Target Presence (T) and Radar Alert (A) are statistically independent. Show your calculations.

Solution

From Section 1: $P(T = \text{Target}) = 0.05$ and $P(A = \text{Alert}) = 0.09$.

$$\text{Expected Independent Joint: } P(T) \times P(A) = 0.05 \times 0.09 = 0.0045$$

$$\text{Actual Observed Joint: } P(T = \text{Target}, A = \text{Alert}) = 0.04$$

Because $0.04 \neq 0.0045$, T and A are **dependent** (not independent). This confirms the radar provides real diagnostic evidence about target presence.

Part 2: Student Independent Practice Solutions

A. Anti-Submarine Warfare (ASW) Acoustic Classification

Scenario: An autonomous sonobuoy array analyzes acoustic contacts in a maritime chokepoint. The system tracks Contact Classification ($C \in \{\text{Hostile}, \text{Biologic}\}$) and Acoustic Frequency Profile ($F \in \{\text{LowFreq}, \text{HighFreq}\}$):

	$F = \text{LowFreq}$	$F = \text{HighFreq}$
$C = \text{Hostile}$	0.12	0.03
$C = \text{Biologic}$	0.18	0.67

1. **Question:** Calculate the marginal probability of detecting a Low-Frequency contact, $P(F = \text{LowFreq})$.

Solution

$$\begin{aligned} P(F = \text{LowFreq}) &= P(\text{Hostile}, \text{LowFreq}) + P(\text{Biologic}, \text{LowFreq}) \\ P(F = \text{LowFreq}) &= 0.12 + 0.18 = \mathbf{0.30} \quad (30\%) \end{aligned}$$

2. **Question:** Calculate the marginal probability that the contact is Hostile, $P(C = \text{Hostile})$.

Solution

$$\begin{aligned} P(C = \text{Hostile}) &= P(\text{Hostile}, \text{LowFreq}) + P(\text{Hostile}, \text{HighFreq}) \\ P(C = \text{Hostile}) &= 0.12 + 0.03 = \mathbf{0.15} \quad (15\%) \end{aligned}$$

3. **Question:** If the sonobuoy detects a Low-Frequency acoustic profile ($F = \text{LowFreq}$), compute the updated conditional probability that the contact is Hostile, $P(C = \text{Hostile} \mid F = \text{LowFreq})$.

Solution

$$\begin{aligned} P(C = \text{Hostile} \mid F = \text{LowFreq}) &= \frac{P(\text{Hostile}, \text{LowFreq})}{P(F = \text{LowFreq})} \\ P(C = \text{Hostile} \mid F = \text{LowFreq}) &= \frac{0.12}{0.30} = \mathbf{0.40} \quad (40\%) \end{aligned}$$

B. Electronic Warfare & Redundant Data Links

Scenario: A command post operates two encrypted communications links: Link 16 (L_1) and SATCOM (L_2). The probability of Link 16 dropping due to adversary jamming is $P(L_1 = \text{Fail}) = 0.20$. The probability of SATCOM dropping due to severe weather is $P(L_2 = \text{Fail}) = 0.10$. Assume the physical causes of failure for the two links are completely independent.

1. **Question:** Compute the joint probability that both communication links fail simultaneously, $P(L_1 = \text{Fail}, L_2 = \text{Fail})$.

Solution

Because link failures are statistically independent:

$$P(L_1 = \text{Fail}, L_2 = \text{Fail}) = P(L_1 = \text{Fail}) \times P(L_2 = \text{Fail})$$

$$P(L_1 = \text{Fail}, L_2 = \text{Fail}) = 0.20 \times 0.10 = \mathbf{0.02} \quad (2\%)$$

- 2. Question:** Compute the probability that at least one communication link remains operational.

Solution

Using the complement rule:

$$P(\text{At least one operational}) = 1 - P(\text{Both fail})$$

$$P(\text{At least one operational}) = 1 - 0.02 = \mathbf{0.98} \quad (98\%)$$

C. Conceptual Analysis: Conditional Independence

Scenario: An autonomous reconnaissance drone carries an Infrared (IR) camera and an Acoustic microphone. Both sensors detect an enemy generator starting.

Solution

Why dependent overall: When the enemy generator starts, it emits both heat and sound simultaneously. Observing an IR alarm increases the probability that the generator is active, which in turn increases the probability that the acoustic sensor will also trigger. Thus, the sensors are correlated/dependent.

Why conditionally independent given the true generator state: Once we directly observe or know with 100% certainty that the generator is active ($G = \text{On}$), the physical heat signature is fixed. Any electrical noise or random sensor artifact in the IR camera provides no mathematical information about random microphone noise in the acoustic sensor:

$$P(\text{IR, Acoustic} \mid G = \text{On}) = P(\text{IR} \mid G = \text{On}) \times P(\text{Acoustic} \mid G = \text{On})$$